



A heterogeneously integrated silicon photonic/lithium niobate travelling wave electro-optic modulator

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Abstract: Silicon photonics is a platform that enables densely integrated photonic components and systems and integration with electronic circuits. Depletion mode modulators designed on this platform suffer from a fundamental frequency response limit due to the mobility of carriers in silicon. Lithium niobate-based modulators have demonstrated high performance, but the material is difficult to process and cannot be easily integrated with other photonic components and electronics. In this manuscript, we simultaneously take advantage of the benefits of silicon photonics and the Pockels effect in lithium niobate by heterogeneously integrating silicon photonic-integrated circuits with thin-film lithium niobate samples. We demonstrate the most CMOS-compatible thin-film lithium niobate modulator to date, which has electro-optic 3 dB bandwidths of 30.6 GHz and half-wave voltages of $6.7 \text{ V} \times \text{cm}$. These modulators are fabricated entirely in CMOS facilities, with the exception of the bonding of a thin-film lithium niobate sample post fabrication, and require no etching of lithium niobate.

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1. Introduction

Electro-optic modulators are utilized as tools to convert electrical signals into optical signals and play vital roles in optical communication networks. Lithium niobate (LiNbO_3) has been utilized as a platform for commercial electro-optic modulators for decades due to its transparency at telecommunication wavelengths and strong second-order nonlinearities (Pockels effect) [1]. Typically, weakly guiding optical waveguides are fabricated on bulk LiNbO_3 by diffusing titanium (Ti) into the Pockels medium. Optical waveguides can also be fabricated through an annealed proton exchange [1]. Metal, usually a thick gold film, is then deposited onto the surface of the LiNbO_3 and patterned to fabricate the RF waveguides which are typically coplanar waveguides (CPWs). These weakly guiding optical waveguides require CPWs with large gaps between electrodes and a large voltage for a given electric field. To counteract this, most commercial devices are 50–150 mm long (2–6 inches), which makes them unsuitable for compact high bandwidth communications and low size, weight, and power (SWaP) applications. Commercially, these types of modulators can reach 3 dB bandwidths of 35 GHz with half-wave voltages of 3.5 V [2]. In these bulk modulators, the electro-optic bandwidth will be limited by the electrical bandwidth of the CPW and the mismatch of velocity between the optical and RF modes.

LiNbO_3 based modulators have seen drastic improvements in performance since the development of thin-film LiNbO_3 (TFLN) technology, or LiNbO_3 on insulator (LNOI), in which a thin layer (hundreds of nanometers) of LiNbO_3 is placed and bonded atop an SiO_2 layer (typically 2 μm thick), which is placed on top of a carrier wafer whose material is LiNbO_3 or silicon. Etching of LiNbO_3 is a difficult task [1], however recently research groups have achieved successful

etching of TFLN resulting in high quality optical waveguides [3–5]. By dry etching this TFLN layer, it is possible to fabricate high index contrast optical waveguides and improve device performance substantially by improving optical waveguide quality and allowing more control over the group index of the optical mode. Devices based on the TFLN platform have shown impressive performance with operating frequencies in the terahertz regime [3] and half-wave voltages as low as 1.4 V [4]. Other groups have implemented various approaches to fabricate photonic devices utilizing the Pockels effect in LiNbO_3 without needing to etch the LiNbO_3 . These approaches typically involve deposition and patterning of silicon nitride (SiN_x) and metal (typically gold) atop of a TFLN wafer [6,7], or fabricating waveguides and bonding TFLN to these waveguides prior to fabricating the rest of the modulator [8–11]. A diagram of the various optical waveguide cross sections for these electro-optic modulators is presented in Fig. 1. While these approaches offer promising performance, it does not support the manufacture of photonic integrated circuits (PICs) utilizing other photonic components (photodiodes, compact spectral filters) and electronics.

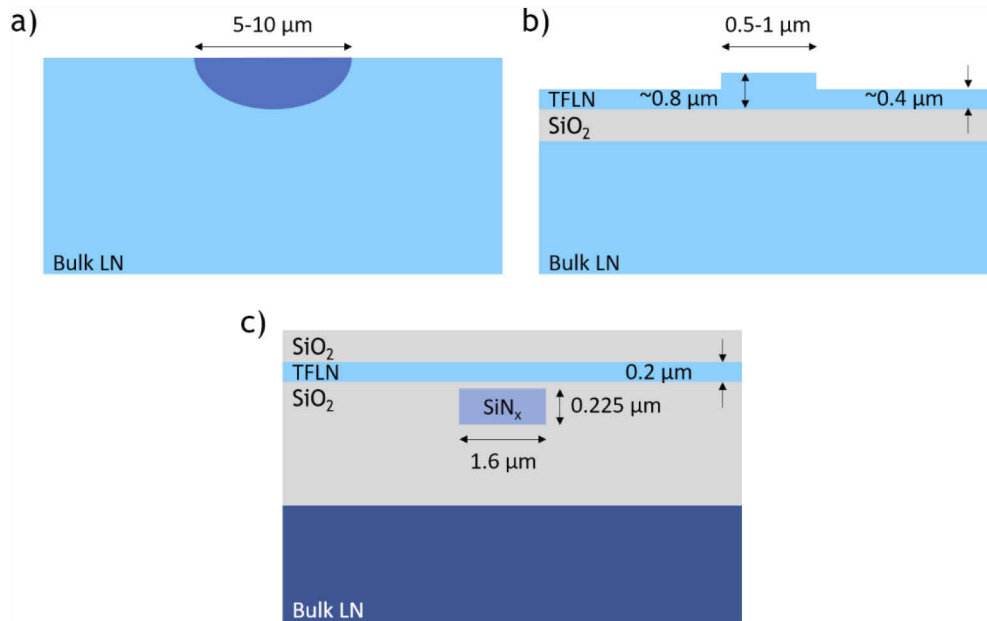


Fig. 1. Optical waveguide configurations utilizing (a) Ti diffused waveguide cores, utilized in [2], (b) etched TFLN waveguide cores, utilized in [3–5], and (c) a waveguide core consisting multiple dielectric materials, which is implemented in [6–8] and in this work.

The silicon photonics (SiP) platform on the other hand has been demonstrated to support complex multi-channel PICs utilizing electro-optic modulators [12,13] and photodiodes. SiP has attractive manufacturing benefits due to the infrastructure developed to build CMOS electronics, and additionally has a monolithic integration capability with these electronics [14]. The ability for integration and ease of manufacturing are advantageous for PICs, but silicon unfortunately has non-ideal optical material properties. Monolithic SiP modulators operate through the plasma dispersion effect (Soref and Bennett model) [15], in which the refractive index of silicon is modulated by changing the carrier density within the waveguide core, and as a consequence, the modulation bandwidth of these modulators is fundamentally limited by the mobility of the carriers within silicon [16]. This method of modulation additionally is nonlinear. Silicon also suffers from two-photon absorption [17] which limits the power handling capability of silicon waveguides. This limit is typically not an issue in low power communications applications

due to the desirability of using low optical power levels [18] but may be a requirement in an RF photonic application [19]. A summary of these modulators is provided in Table 1.

Table 1. Comparison of reported monolithic and hybrid electro-optic modulators.

Platform	Electro-optic bandwidth (GHz)	Half-wave voltage length product (V×cm)	Reference
Bulk LiNbO₃	35	>10	[2]
Integrated LiNbO₃	500 ^a	3.8	[3]
Integrated LiNbO₃	100	2.2	[4]
Integrated LiNbO₃	110 ^a	9.4	[5]
SiP	30	2.6	[9]
SiP	34	0.74	[10]
Hybrid SiN_x/TFLN	8	3	[6]
Hybrid SiN_x/TFLN	33	3.1	[7]
Hybrid SiP/TFLN	>106	6.7	[8]
Hybrid SiN_x/TFLN	30.6	6.67	This work

^aOperation up to these frequencies.

The goal of this work is to develop electro-optic modulators that utilize the material properties of LiNbO₃ while simultaneously leveraging the manufacturing benefits of SiP. We propose a SiP/TFLN heterogeneously integrated platform which consists of fabricating SiP PICs (SiPICs) using Sandia's CMOS fabrication facilities and bonding a sample diced from a TFLN wafer to the smoothed surface of the SiPIC (see Fig. 2). Aside from utilizing the benefits from the monolithic TFLN and SiP platforms, this approach also does not require any etching of LiNbO₃, which while possible, is a difficult process. SiN_x waveguides [20,21] are also incorporated in this platform which can provide lower fiber to chip coupling losses and remove the limitation on optical power handling capacity using silicon waveguides, while still providing the capability of monolithic integration with CMOS electronics. This approach enables simplified fabrication of TFLN devices and systems and addresses the material limitations of silicon with respect to modulation and power handling capacity. Other groups have built devices leveraging the optical material properties of LiNbO₃ without the need to etch the material [6–11], but these methods all require processing outside of a CMOS foundry.

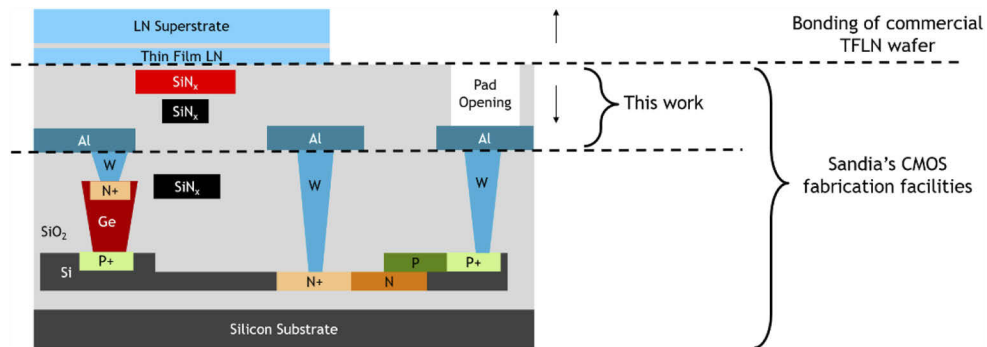


Fig. 2. A cross section of the proposed heterogeneously integrated SiP/TFLN platform utilizing standard SiP CMOS compatible materials bonded to TFLN chips.

In this manuscript, we discuss the design, fabrication and characterization of a heterogeneously integrated SiP/TFLN travelling wave modulator. The fabricated devices in this work only utilize the materials necessary to realize modulation which are aluminum (Al) and two SiN_x waveguide

layers encased in SiO₂ bonded to a TFLN chip which is diced from commercially obtained TFLN wafers (Fig. 2). The methodology implemented in our work allows for complete PIC fabrication using CMOS facilities with the exception of dicing and bonding of a TFLN chip. Because the RF and optical structures are fabricated completely in CMOS facilities, these TFLN modulators are the most CMOS compatible TFLN modulators to date, to the best of the authors' knowledge. Modulators designed on this platform do not suffer from the fundamental bandwidth limitation of depletion-mode SiP modulators or the two-photon absorption phenomena in silicon, which would otherwise be an effect in silicon waveguides or hybrid waveguides using silicon as a core material. The fabricated 0.5 cm long modulators have electro-optic 3 dB bandwidths of 30.55 GHz and half-wave voltages of 6.67 V×cm.

2. Modulator design and fabrication

The finalized hybrid device is an amplitude modulator in the Mach-Zehnder configuration. It consists of two phase modulators as the arms of the Mach-Zehnder modulator (MZM) in a push-pull configuration to reduce the half-wave voltage, V_π . Each of the phase modulators that make up the arms of the Mach-Zehnder modulator (MZM) consist of two layers of SiN_x waveguides; one of which serves to route the guided optical mode beneath the TFLN region to avoid optical loss to reflection at the air/TFLN interface, and the other is a hybrid waveguide whose guided mode resides in the TFLN as well as the SiN_x in order to take advantage of the Pockels effect in the TFLN while maintaining a guided mode. Light is transferred between the top and bottom SiN_x waveguide layers via two overlaying 250 μm linear tapers, with a minimum width of 0.2 μm and a maximum width of 1.2 μm. The top layer waveguide additionally tapers out from 1.2 μm to 1.6 μm over a 100 μm distance. A schematic of the two-layer SiN_x waveguides used in the arms of the MZM, along with the evolution of a guided mode through such a structure, is displayed in Fig. 3. The two-layer waveguide structure is necessary to simultaneously achieve optimal optical transmission across the device and high overlap of the optical mode with TFLN. A single waveguide layer device would result in either poor optical transmission due to losses at the air/TFLN interface, as indicated in Fig. 3, or poor modal overlap with the TFLN, resulting in non-ideal half-wave voltages.

These devices are fabricated using Sandia's microsystems engineering, science and applications (MESA) complex. The starting materials are 3 μm of thermally grown SiO₂ atop of high resistivity (> 1000 Ω×cm) silicon wafers, which are used to minimize attenuation of the RF mode (due to dielectric loss). A metal stack is deposited on top of the thermally grown SiO₂, which consists of Ti/TiN/Al with a total thickness of 870 nm is patterned to form the CPWs. High density plasma (HDP) oxide is deposited and a chemical-mechanical planarization (CMP) step is utilized to provide a flat surface a target of 250 nm above the metal circuit. Then, a 300 nm thick silicon rich plasma enhanced chemical vapor deposition (PECVD) SiN_x film [21] with a material index of 2.11 (at 1546 nm) is deposited and patterned to form the bottom waveguide layers. A second layer of HDP oxide is deposited and another CMP step is utilized to achieve a target thickness of 250 nm of SiO₂ above the bottom waveguide layer in an effort to improve coupling efficiencies between top and bottom waveguide layers. A second PECVD SiN_x film, with a material index of 1.91 (at 1546 nm) and a thickness of 225 nm is deposited atop this smooth SiO₂ surface and patterned before another HDP oxide deposition. The SiO₂ layer above the top waveguide layer is planarized using a final CMP step such that there is a targeted SiO₂ thickness of < 100 nm above the top waveguide layer, however this is difficult to control in the CMP process and the measured thickness of SiO₂ above the top waveguide layer is 123 nm (via analyzing an SEM of a device cross section). This is a nominal value for this SiO₂ thickness however there is expected to be variation throughout the wafer due to CMP variations. Minimizing the amount of SiO₂ above the top waveguide layer helps ensure that more of the guided mode will reside in the TFLN as well as the SiN_x, which directly improves the efficiency of the modulator (i. e. lowers V_π).

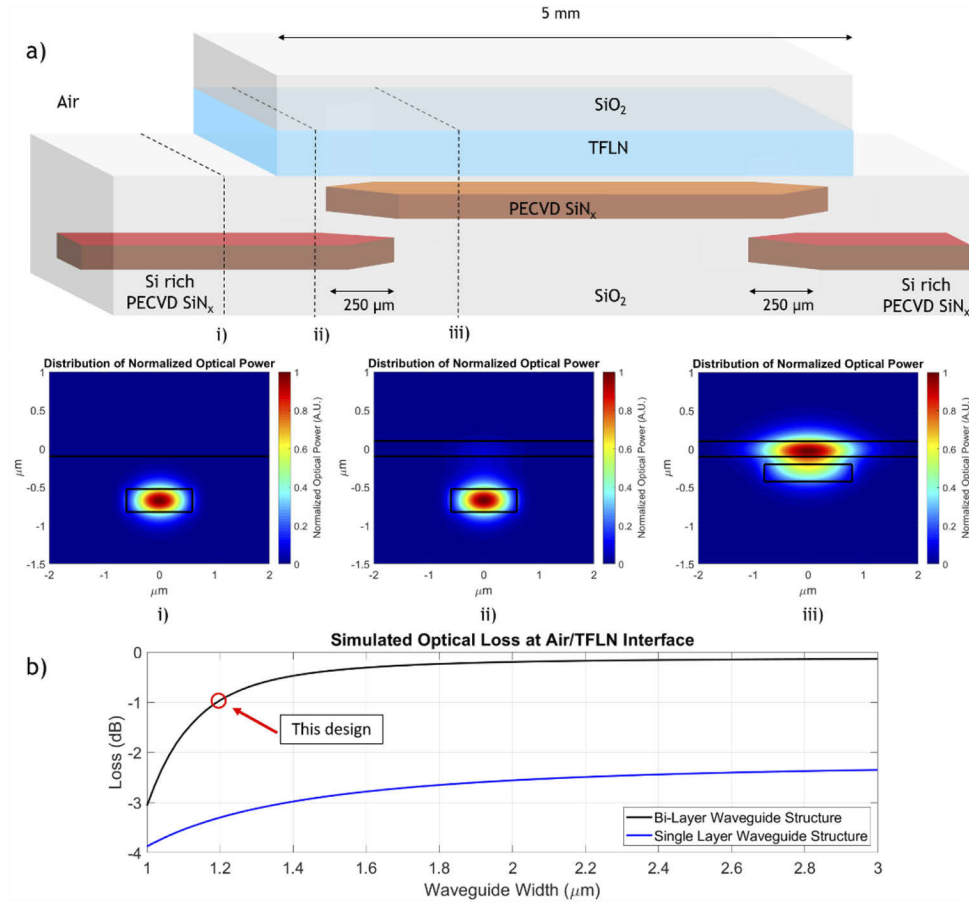


Fig. 3. (a) Perspective schematic (not to scale) of the heterogeneously integrated SiP/TFLN phase modulators and corresponding evolution of optical power distribution through the structure, where light enters through the i) bottom Si-rich SiN_x waveguide, ii) propagates underneath the TFLN/air interface, and iii) couples up to the hybrid SiN_x/TFLN waveguide using 250 μm long tapers in each waveguide layer. Light couples out of the phase modulator region in the reverse fashion. (b) Simulated optical loss at the air/TFLN interface for the implemented bi-layer waveguide structure and a single waveguide structure as a function of waveguide width.

Openings in the oxide are patterned and etched for access to the fabricated RF structures. This planarized SiO₂ surface will then be the bonding surface in which the diced X-cut TFLN chips will be bonded to.

The bonding of the TFLN chips to the SiP samples is accomplished at low temperatures, where the bonding process is realized through van der Waals forces [8]. The two bonding surfaces are the TFLN surface, and the planarized SiO₂ surface of the SiP sample. The TFLN bonding samples are diced from commercially obtained TFLN wafers (NanoLN) and consist a 500 μm thick handle wafer consisting of X-cut LiNbO₃ with a 2 μm oxide layer and a nominal 200 nm thick TFLN layer (also X-cut). In order to achieve quality bonds, the surfaces to be bonded must be very smooth. The bonding surface of the silicon photonic sample is planarized using CMP steps to have a targeted root-mean square (RMS) roughness of less than 10 nm and is nominally 6 nm. There are no polishing or planarization steps performed on the TFLN chips. The two samples

are cleaned using an SC1 clean ($\text{H}_2\text{O}:\text{H}_2\text{O}_2:\text{NH}_4\text{OH}$, 60:4:1 at 40°C with a 75 W megasonic applied for 120 seconds), proceeded by a 60 second O_2 plasma activation. This process removes hydrocarbon particles while increasing silanol Si-OH group densities at the bonding surfaces yielding a larger number of bonding sites. Once each bonding surface is cleaned, the bond is initiated by pressing the two samples together with a force of 500 Newtons for twelve hours while applying a thermal anneal of 50°C to encourage covalent bonding. Relatively low bonding temperatures are used here to avoid damage to the samples which would be expected due to the mismatch of coefficients of thermal expansion (CTE) between LiNbO_3 and SiO_2 . The limits of bonding temperature and bonding strength were not explored in this work and are the topic of future investigation. A schematic of the modulator and a cross section of the device prior to bonding with labelled dimensions, as well as a photograph of the fabricated device is presented in Fig. 4.

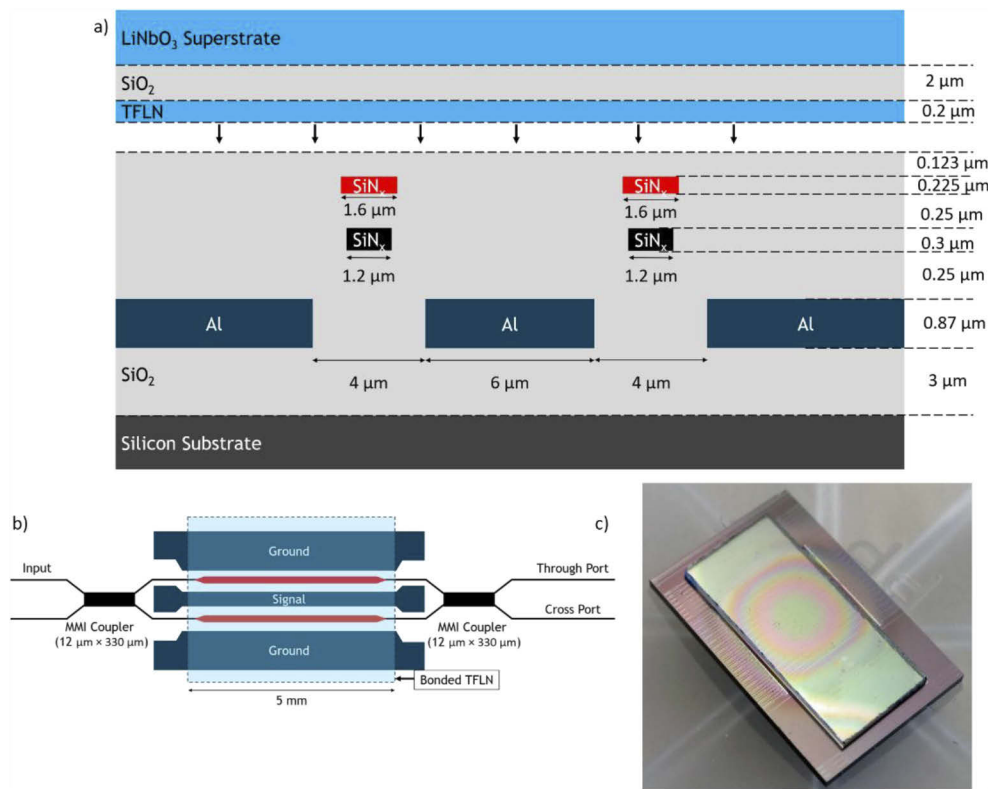


Fig. 4. (a) Cross section of the final, pre-bond cross-section of the MZM with dimensions labelled (not to scale), (b) Schematic of the bonded modulator detailing the signal and ground traces of the CPW, and the hybrid SiN_x/TFLN phase modulators in red, and (c) a photograph of the TFLN chip bonded to the SiP sample. The concentric fringes are due variations in SiO_2 arising from variations in the CMP process, while the diagonal fringes on the bottom left of the transparent TFLN sample are due to the sample not being completely bonded.

The fringes seen in the photograph in Fig. 4 arise from variations in the thickness in SiO_2 which are a result of CMP variations. The planar structures under the surface of the SiP sample can be designed to have a more uniform distribution of material density, as well as utilization of higher quality CMP tools, to achieve more uniform SiO_2 thicknesses between the TFLN and topmost SiN_x waveguides. For reference, the metal density is nearly 100% near the center of

the fringes where the CPWs are located, and outside of this area the metal density is smaller ($\sim 50\%$). The fringes can also be expected due to variations in bond quality, where a thin layer of air is present between TFLN and SiO_2 . The thickness of SiO_2 between the optical waveguides and TFLN is difficult to control using CMP and heavily impacts the characteristics of the optical waveguide and affects the distribution of the RF field (to a much lesser extent).

Two phase modulators make up the arms of an MZM, between the signal and ground traces of a coplanar waveguide (CPW) to realize amplitude modulation. The 3 dB splitters at the input and output of the MZM are fabricated in the bottom routing SiN_x waveguide layer. To realize broadband optical operation, multi-mode interference (MMI) devices are implemented. The dimensions of the 4-port MMI 3 dB couplers are $330 \mu\text{m} \times 12 \mu\text{m}$ with waveguide tapers from $1.2 \mu\text{m}$ to $4 \mu\text{m}$ into the MMI slab to reduce optical loss. The CPW used in the MZM has a gap (G_{CPW}) of $4 \mu\text{m}$ between the signal and ground traces with a signal trace width (W_{CPW}) of $6 \mu\text{m}$. The ground planes of the CPW are $130 \mu\text{m}$ wide. In Fig. 4, the electrodes apply equal amplitude fields in opposite directions to each optical waveguide simultaneously. Because the electro-optic effect is dependent on the direction of the field, the two phase shifters act in a complementary manner often referred to as push-pull operation, where one waveguide experiences phase shift $+\Delta\varphi$, while the other experiences a phase shift of $-\Delta\varphi$.

$$V_\pi = \frac{1}{L} \frac{n_{eff} \lambda_0 G_{CPW}}{2n_e^2 r_{33} \Gamma} \quad (1a)$$

$$\Gamma = \frac{G_{CPW}}{V} \frac{\int |E_o|^2 \times E_{RF}^{TE} ds}{\int |E_o|^2 ds} \quad (1b)$$

The half wave voltage, V_π , for a push-pull configuration MZM can be calculated from knowledge of the phase modulator geometry using Eq. (1) where λ_0 is the optical wavelength of operation, L is the length of the phase modulator, n_{eff} is the effective index of the guided mode (simulated to be 1.757 at 1550 nm), n_e is the index of refraction in LiNbO_3 along the extraordinary axis (2.21 at 1550 nm), r_{33} is the electro-optic coefficient in the TE direction (31 pm/V in LiNbO_3), and Γ is the overlap of the optical (E_o) mode and RF mode along the optical axis (E_{RF}^{TE}) of the LiNbO_3 within the Pockels material [22] with an electrical potential V between the center and ground conductors of the CPW. The factor of two in Eq. (1) is due to the push-pull configuration. This overlap factor is predicted by simulating the modal distribution of the guided optical mode using finite-difference time domain (FDTD) simulations, and the RF modal distribution using finite element method (FEM) simulations. Simulations are performed with 123 nm of SiO_2 above the top SiN_x waveguide layer, which is the mean measured value of this SiO_2 acquired via ellipsometry. Using Eq. (1), the half-wave voltage of this device is predicted to be $4.88 \text{ V}\times\text{cm}$, and the overlap between RF and optical modes, Γ , is found to be 19.03% using simulation results. The simulated hybrid structure consists of a 200 nm thick slab of LiNbO_3 , with a rectangular SiN_x waveguide core ($1.6 \mu\text{m} \times 0.225 \mu\text{m}$) with 123 nm of SiO_2 between the LiNbO_3 and SiN_x , encased in SiO_2 as the rest of the waveguide cladding.

The interaction between the guided optical mode and RF field distribution within the electro-optic material is approximated as Eq. (1b). The mechanism for altering the refractive index is the strength of the RF field within this material, while the convention for efficiency is driving voltage, as seen in Eq. (1a). To address this, the factor of G_{CPW}/V is included in the expression for the overlap factor Γ and is an approximate value of the electric field in the electro-optic material resulting from applying electric potential V between the signal and ground electrodes of the CPW.

$$v_{RF}(\omega, z, t) = e^{j\omega t} (V^+ e^{-\gamma_{RF}(\omega)z} + V^- e^{\gamma_{RF}(\omega)z}) \quad (2a)$$

$$\gamma_{RF}(\omega) = \alpha_{RF}(\omega) + j\omega \sqrt{\epsilon_{RF}(\omega)}/c \quad (2b)$$

$$V^+ = \frac{1}{1 - \Gamma_L \Gamma_G \exp(-2\gamma_{RF}L)} \frac{Z_0}{Z_0 + Z_G} V_A \quad (2c)$$

$$V^- = \frac{\Gamma_L \exp(-2\gamma_{RF}L)}{1 - \Gamma_L \Gamma_G \exp(-2\gamma_{RF}L)} \frac{Z_0}{Z_0 + Z_G} V_A \quad (2d)$$

The CPW is designed to have a characteristic impedance, Z_0 , of 50 Ω , minimal RF attenuation and the matching of the velocity of the RF and optical mode to achieve optimal frequency response. These constraints are realized through numerical methods (FEM simulations), and these simulated constituent parameters for the CPW used in this MZM design are plotted in Fig. 5. Additional curves are included for electrode gaps of 3.5 μm and 4.5 μm to illustrate how sensitive the device is to this parameter. Because the RF waveguide is fabricated using CMOS compatible materials and dimensions, the CPWs consist of aluminum which is 0.87 μm thick. The RF waveguides then are very thin, and RF attenuation will always be a limiting factor in the performance of devices built on this platform. The voltage of a travelling RF wave, v_{RF} , at time t , position z , and at angular frequency ω on a CPW consists counterpropagating modes with phase velocity $c/\epsilon_{RF}^{1/2}$ as follows Eq. (2), where c is the speed of light, ϵ_{RF} is the effective permittivity of the RF mode, γ_{RF} is the propagation constant of the RF mode, and α_{RF} is the attenuation factor. The constants V^+ and V^- are the amplitude of the forward and backward propagating RF modes, and depend on the reflections of the RF modes, Γ_G and Γ_L at the generator and load, with respective impedances Z_G and Z_L , and the open-circuit voltage at the generator, V_A .

$$v_{RF}(\omega, z, t_0 + zn_g/c) = e^{i\omega t_0} [V^+ e^{(n_g/c - \gamma_{RF}(\omega))z} + V^- e^{(n_g/c + \gamma_{RF}(\omega))z}] \quad (3a)$$

$$\Delta\phi(\omega, t_0) = \frac{2\pi}{\lambda_o} \int_0^L \Delta n(\omega, z, t_0 + zn_g/c) dz = \Delta\Phi(\omega) e^{i\omega t_0} \quad (3b)$$

$$\Delta n\left(\omega, z, t_0 + z \frac{n_g}{c}\right) = \frac{n_e^4 r_{33}}{2n_{eff}} \frac{\Gamma_L}{G_{CPW}} v_{RF}\left(\omega, z, t_0 + z \frac{n_g}{c}\right) \quad (3c)$$

$$m_{Elec}(\omega) = \left| \frac{\Delta\Phi(\omega)}{\Delta\Phi(0)} \right|^2 \quad (3d)$$

To further formalize the frequency response of the modulator, the phase variation of the optical mode can be associated with Eq. (2) by parametrizing t as a function of position and optical group index n_g [23] such that $t(z) = t_0 + zn_g/c$, where t_0 is the time the optical mode enters the modulation region (Eq. (3a)). Because LiNbO₃ is a Pockels medium, the change in refractive index is linear to an applied electrical field (or equivalently, voltage), such that $\Delta n \propto v_{RF}$. The bandwidth of a Pockels medium electro-optic modulator will be dictated by the amount phase difference between the optical modes in each arm of the modulator, which is directly proportional to the voltage on the CPW. The phase variation along the axis of propagation, as a function of modulation frequency, as seen by the optical mode, is then given as Eq. (3b). The electrical bandwidth $m_{Elec}(\omega)$ of the modulator is finally calculated using these parameters in Eq. (3d).

The bandwidth of Pockels medium electro-optic modulators is governed by the electrical attenuation of the voltage on the CPW and the mismatch the RF phase velocity and optical group velocity. This velocity mismatch can be intuitively understood by considering the phase fronts of the optical and RF modes periodically becoming in and out of phase relative to one another, which is compounded in modulators utilizing large lengths, and leads to a reduction in total phase change accumulation at the end of the modulation region. The effect of electrical attenuation on bandwidth is trivial and is attributed to less voltage (and therefore less change in refractive index) in the modulation region as the attenuation increases with modulation frequency. It is worth noting that this formalism is valid for a phase modulator, and in our push-pull configuration, two phase modulators are utilized such that the change in optical phase in each arm is opposite in polarity from one another and provides an overall phase difference between arms of $2\Delta\phi$. While the push-pull configuration does not provide improvements to frequency response, it directly improves the driving efficiency by a factor of two, and is a common method used to reduce the half-wave voltage V_π .

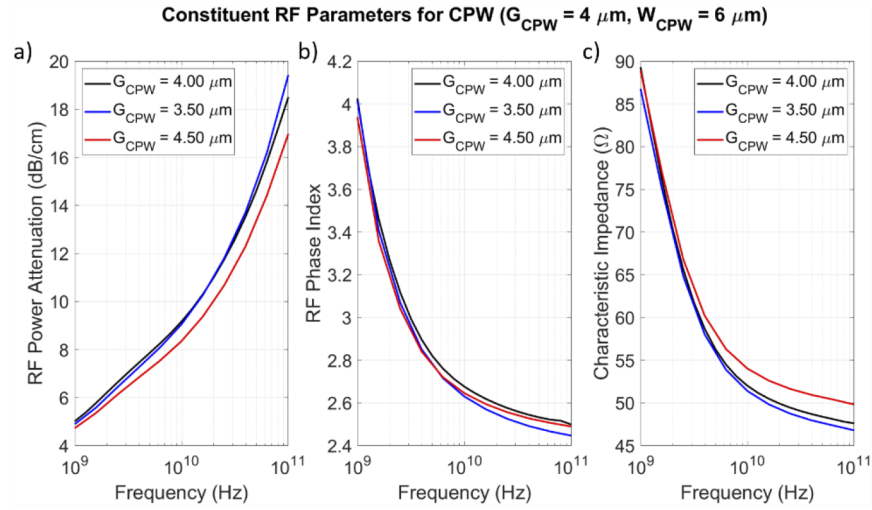


Fig. 5. The simulated (a) RF attenuation, (b) phase index, and (c) characteristic impedance for the CPW used in the heterogeneously integrated MZM as a function of frequency for the designed gap of $4 \pm 0.5 \mu\text{m}$.

It is difficult to control the thickness of SiO_2 between the TFLN and SiN_x within the modulation region of the device through the CMP process, and because of this, the device needs to be robustly designed such that it will function regardless of this variation. This distance is very small compared to RF wavelengths and does not appreciably impact the performance of the CPW. However, the optical mode is quite sensitive to this thickness, and the effective/group index of the guided mode will change with the amount of SiO_2 between each of the hybrid waveguide cores. Additionally, the overlap of the optical mode with the TFLN will change with this thickness as well. To realize a robust design for the modulator, the predicted electro-optic bandwidth and half-wave voltage are calculated over a range of SiO_2 thickness between TFLN

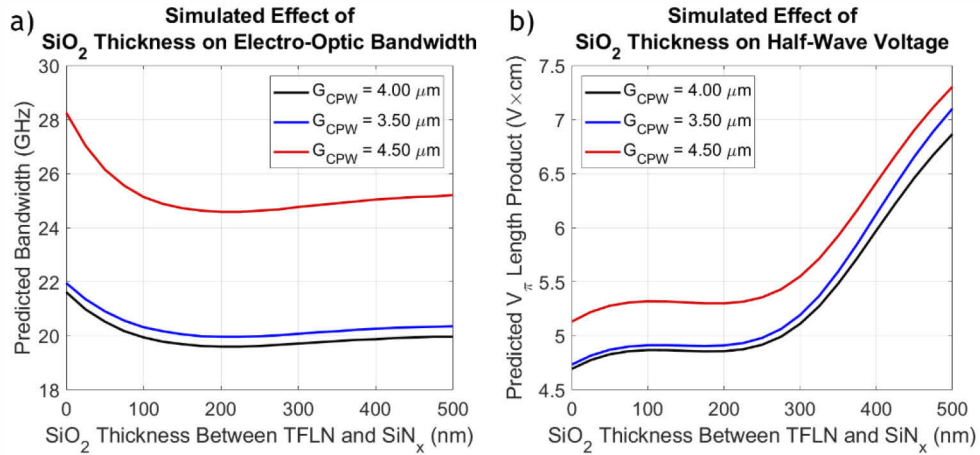


Fig. 6. The simulated sensitivity to SiO_2 thickness between TFLN and SiN_x for the (a) electro-optic bandwidth and (b) half-wave voltage of the heterogeneously integrated MZM with an electrode gap of $4 \pm 0.5 \mu\text{m}$, where a clear tradeoff between bandwidth and half-wave voltage can be seen as the electrode gap is increased.

and SiN_x are plotted in Fig. 6. These simulations implement the dimensions provided in Fig. 4 for the designed CPW gap of $4 \pm 0.5 \mu\text{m}$ to show how sensitive this modulator design is to this parameter. Additionally, the material index of the TFLN is higher than that of the SiN_x , so the hybrid waveguide must be designed carefully to ensure that the fundamental mode of the structure is not the slab mode of the TFLN. This is an important design consideration and neglect can result in light coupling into the TFLN slab mode, rendering the device inoperable.

3. Experimental results

The modulator was designed with an extra length of passive waveguide (bottom routing SiN_x) in one arm to enable simple measurement of the MMI balance and modulator extinction across a wavelength span as shown in Fig. 7. The modulator is biased at quadrature by setting the optical wavelength of the laser source such that the output light is 3 dB less from the maximum optical transmission in the MZM spectra, which is shown in Fig. 7. The approach eliminates a bias tee that would be used to set the bias point in a balanced device; in a broadband optical data application, one would choose the bias tee approach. There is a weak absorption in SiN_x around 1520 nm [21], and results in the curvature in the spectra of the MZM. At peak transmission around 1550 nm, the total insertion loss through the cross port of the MZM is 13 dB. Light is injected into the device and collected using lensed fibers with a $3.5 \mu\text{m}$ spot size.

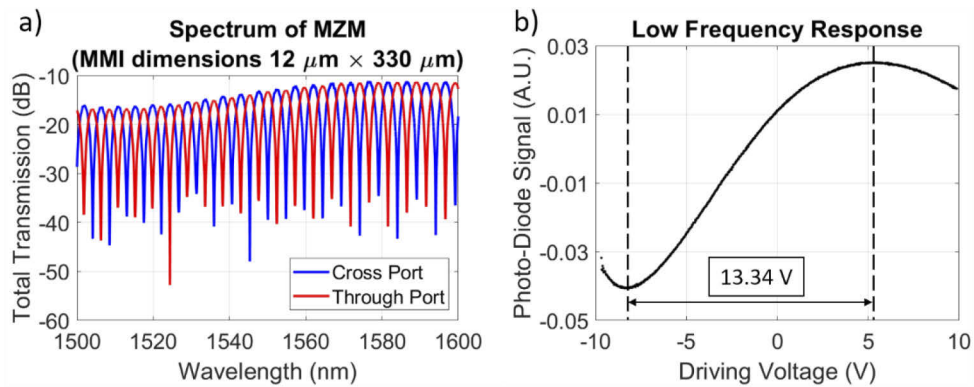


Fig. 7. (a) Spectra of both outputs of the un-bias MZM with near constant splitting ratio and extinction ratios above 20 dB from 1500 nm to 1600 nm and (b) low frequency half-wave voltage (V_π) results measured by applying a 50 kHz sawtooth signal to the 0.5 cm long MZM and measuring the electro-optic response using an amplified photodiode.

Due to the complexity of the structure, there are numerous sources of loss which arise from a mismatch between the fiber mode and planar waveguide mode, propagation loss of each waveguide, mismatch of the mode on either side of the TFLN chip edge, optical loss through the $250 \mu\text{m}$ long vertical couplers, MMI coupler loss, and the bent waveguides routing light into the MMI devices. The simulated loss for each of these components, along with measured loss (where feasible) are summarized in Table 2. The simulations implement the dimensions featured in Fig. 4. These simulations likely underestimate loss for these components due to non-ideal experimental conditions (fabrication nonuniformities, roughness at the bonding interface and non-uniform SiO_2 thickness above the topmost SiN_x waveguide due to CMP variations) that are difficult to capture in simulation, resulting in the disagreement between measurements and simulation. Insertion loss of the MZM's can be improved through careful component design (MMI, waveguides at the air/TFLN interface, bends). Better design for the planar waveguide dimensions routing light into the chip from the fiber can significantly improve optical transmission, although this is a difficult

problem due to the asymmetry of the structure at the end of the chip combined with the relatively large size of the fiber mode.

Table 2. Measured and simulated loss for each MZM element.

MZM element	Simulated Loss per Element (dB)	Total Measured Insertion Loss (dB)
Fiber-to-chip interface	-2.34	N/A
Propagation loss	-1.28 ^a [21]	N/A
Air/TFLN interface	-0.96	N/A
Vertical coupler	-0.15	N/A
MMI coupler	-0.55	N/A
Bending loss	-0.49 ^a	N/A
Passive waveguide (no TFLN interaction)	-5.96	-7.75
Phase modulator	-8.18	-9.84
Complete modulator	-9.77	-13.4

^aTotal contribution to insertion loss.

The half-wave voltage is measured through applying a low frequency (50 kHz) sawtooth wave to the heterogeneously integrated MZM. Light is injected into the modulator from a tunable laser biased at 1551.64 nm at power levels of 6 dBm, and the modulator output is measured using an amplified InGaAs detector. The measured value of V_π is 6.67 V×cm for this MZM with an electrode gap of 4 μ m. The disagreement between this measurement and Eq. (1) arises from fabrication uncertainties which make modelling difficult, particularly the effective index of the guided mode and the overlap factor between optical and RF modes. This can be directly attributed to non-ideal bonding surfaces resulting in variations at this surface along the length of the phase modulators. Because the MMI couplers are broadband photonic components, the fabricated MZM can be biased at quadrature at using several values of wavelengths from 1500 nm to 1600 nm. For all measurements reported here, the bias wavelength is arbitrarily chosen to be 1551.64 nm but can be operated at any wavelength 3 dB down from maximum power output.

The electro-optic bandwidth is measured in a similar way. The modulator is driven by one port of a vector-network analyzer (VNA) while biased at quadrature by tuning the optical wavelength to 1551.64 nm. The optical output of the modulator is amplified using an erbium doped fiber amplifier (EDFA) to overcome insertion loss in the device (~13 dB total loss at 1550 nm using lensed fibers) and filtered using a 1 nm bandpass filter before being routed into a high-speed photodiode (3 dB bandwidth of 42 GHz) whose electrical output is then fed into the second port of the VNA. The bandwidth of the CPW used in the modulator, along with the electro-optic bandwidth of each modulator, is plotted in Fig. 8, where a 3 dB electro optic bandwidth of above 30 GHz is observed. Integration of Eq. (3b) is straight forward if the group index of the guided optical mode is constant throughout the phase modulation region. This is not strictly true in the case of this modulator due to the changing waveguide structure (vertical transitions, routing bottom waveguides), but serves as a first order approximation. Electrical bandwidth of the CPW is measured independently of the MZM electro-optic bandwidth and plotted against simulated bandwidth in Fig. 8. There are RF losses in the measurement unaccounted for in simulation due to a sheet charge present between the high-resistivity silicon substrate and SiO₂ [24], which may account for the disagreement between simulation and measurement. The simulated group index of the guided optical mode in the phase modulator region is found to be 2.051, and, along with the simulated frequency dependent CPW parameters, Eq. (3d) may be evaluated as shown in Fig. 8. The disagreement between measured frequency response is explained through variation in the SiO₂ thickness between LiNbO₃ and SiN_x along the length of the phase modulator, which results in variation of the optical group index along the length of the phase modulator. The model

may be improved through more accurately simulating the optical group index along the phase modulator and calculating Eq. (3) numerically. The variation in the bond between the SiO₂ and LiNbO₃ along the phase modulators cause this uncertainty, similar to the issue with modelling the half-wave voltage.

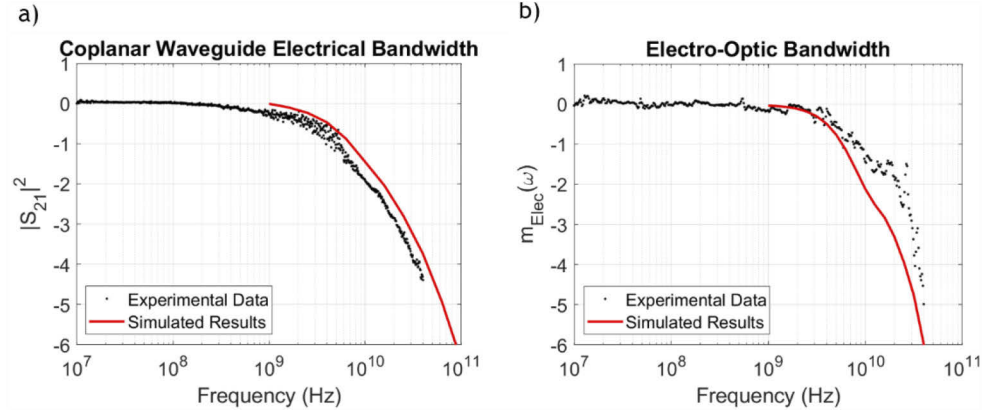


Fig. 8. The electrical 3 dB bandwidth of the CPW used in the travelling wave modulator is 20 GHz and shows a 6 dB bandwidth above 40 GHz (a), the 3 dB electro-optic bandwidth of the travelling wave modulator is 30 GHz, and experimental results are plotted against modelled frequency response using Eq. (3b).

To assess the linearity of the hybrid SiP/TFLN photonic modulator in an RF photonic link, the spurious-free dynamic range (SFDR) is also measured. The result of the measurement is seen in Fig. 9. SFDR is measured by driving the modulator with two RF signals ($f_1 = 1$ GHz, and $f_2 = f_1 + \delta f$, where $\delta f = 100$ MHz) with amplitudes V_0 . The optical wavelength is again set to 1551.64 nm to bias the MZM at quadrature and the laser output power is set to 12 dBm. The modulated optical signal is fed into our high-speed photodiode and the electrical signal is recorded using an RF spectrum analyzer with a normalized noise floor of -147.89 dBm/Hz. It is worth noting here that the noise floor here is normalized to the bandwidth of the measurement. During any measurement with a spectrum analyzer, power is measured over some bandwidth, and because of this, RF noise power (in units of dBm), increases with larger resolution bandwidths. To isolate the measured noise floor from spectrum analyzer measurements, such as dynamic range, the noise floor is normalized to the resolution bandwidth in the measurement, and it is this normalized noise floor that is typically used in analyzing RF photonic links. This subtlety is illustrated Fig. 9, where the noise floor in the measurement can be seen to be roughly -110 dBm while the normalized noise floor is -147.89 dBm/Hz. The SFDR for modulators based on various platforms is summarized in Table 3.

$$SFDR = \left(\frac{P_{Out}}{N_{Out}} \right)^{2/3} \quad (4)$$

The fundamental response of the modulator (f_1, f_2) along with the third order intermodulation distortion (IMD3) terms ($2f_2 - f_1, 2f_1 - f_2$) are recorded as a function of RF input power, as seen in Fig. 9. On a logarithmic scale (both the driving signal and measured response), the fundamental RF tones scale linearly with the input RF power at a slope of unity, while the IMD3 terms scale linearly with a slope of three. By extrapolating upon these experimentally collected values, the intercepts of these curves are found to be (39.09 dBm, 16.91 dBm). Using the output power of the intercept point ($P_{Out} = 16.91$ dBm), and the measured noise floor of the RF photonic link ($N_{Out} = -147.89$ dBm/Hz), the SFDR of our link using our MZM with CPW gap of $4 \mu\text{m}$ is

measured to be $94.162 \text{ dB} \times \text{Hz}^{2/3}$ using Eq. (4).

$$I_d(t) = P_{Las} g_{amp} T_{MZM} \cos^2 \left[\phi_{Quad} - \pi \frac{V_0 (\sin 2\pi f_1 t + \sin 2\pi f_2 t)}{V_\pi} \right] \mathcal{R} \quad (5)$$

$$N_{Out} = (1 + g_{link}) k_B T_0 + 2q I_d R_L + \left[RIN_{Las} + \frac{2h\nu}{P_{Las}} \frac{nf_{amp}}{T_{MZM}} \right] I_d^2 R_L \quad (6)$$

The measured SFDR shows good agreement with the modelled link in Eq. (5), which utilizes the transfer functions of the components in the link. The optical power at the photodiode can be expressed using the transfer function of the MZM biased at quadrature with phase ϕ_{Quad} , whose amplitude is equivalent to the linear product of the laser power (P_{Las} , 12 dBm), insertion loss of the MZM (T_{MZM} , -13 dB), and gain of the EDFA utilized between the MZM and the photodiode (g_{amp} , 12 dB). This is converted to current in the photodiode (I_d) by multiplying the optical power by the responsivity ($\mathcal{R}$, 0.6 A/W). This time domain current is easily converted to the output power of the link using Ohm's law and 50 Ω load resistance R_L , and the fundamental and third order distortion terms can be found by using the Fourier transform and analyzing the frequency domain power output. The output noise, N_{Out} , in units of W/Hz, is calculated using specified values of noise contributing components in the link as seen in Eq. (6), whose first term arises from thermal noise, second term from shot noise, and final term from relative intensity noise (RIN) in the laser and EDFA. In Eq. (6), k_b is the Boltzmann constant (1.38×10^{-23} J/K), T_0 is room temperature (290 K), g_{link} is the link gain (-38.13 dB), q is the electron charge (C), RIN_{Las} is the laser noise (-150 dBc/Hz), $h\nu$ is the photon energy at 1550 nm (J), and nf_{amp} is the noise figure of the EDFA (5 dB). The SFDR is then calculated by analyzing the range of RF power that the link can be operated over while such that the power carried by the 3rd order frequencies (IMD3 terms) remains below the noise floor of the link, which is an alternative but equivalent definition to Eq. (4). By using the specified noise figures (NF's) of each component, we determine dominating noise arises from the relative intensity noise (RIN) of our tunable laser, which is calculated to be -151.48 dBm/Hz using Eq. (6). The excess experimentally measured noise power is attributed to the presence of noise in the spectrum analyzer used in the experiment. SFDR can be improved through increasing the difference in the interpolated third

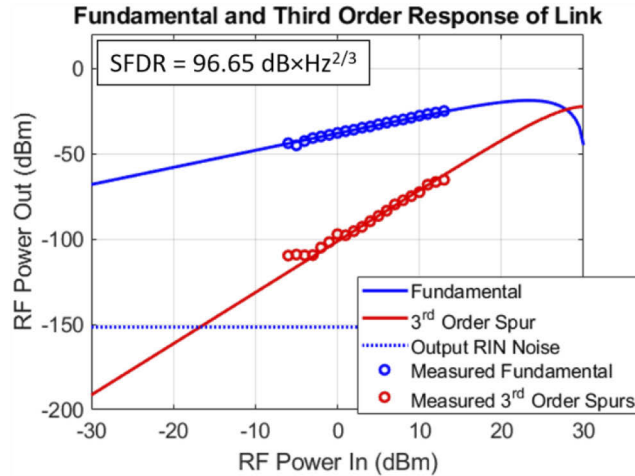


Fig. 9. Measured and modelled fundamental and third order intermodulation distortion (IMD3) response of the hybrid TFLN modulator at 1 GHz. The noise floor (in units of dBm) can be clearly seen around -110 dBm, while the normalized noise floor (in units of dBm/Hz) is plotted as blue dashed line and is dominated by RIN.

order intercept point and the noise floor of the link. In practice, this can be accomplished through improving optical loss in the link (improving device insertion loss for example) or through utilizing differential detection schemes [25]. These scenarios are modelled using the parameters of the experimental environment discussed here and summarized in Table 4.

Table 3. Reported SFDR for various electro-optic modulators.

Modulator type	Spurious free dynamic range (dB Hz ^{2/3})	Notes	Reference
Hybrid SiN_x/TFLN MZM	97.3 (at 1 GHz)	SiN deposited on TFLN	[7]
Hybrid Si/TFLN ring	98.1 (at 1 GHz)	Bonded TFLN to Si ring	[11]
Bulk LiNbO₃	94.7 (at 1 GHz)	Commercial modulator	[11]
Hybrid SiN_x/TFLN MZM	96.65 (at 1 GHz)	Bonded TFLN to SiN _x MZM	This work

Table 4. Predicted dynamic range and noise figure under various conditions

Modelled SFDR (dB×Hz ^{2/3})	Modelled Link Noise Figure (dB)	Conditions
96.7	60.7, RIN limited	Measured experimental values
100.7	54.6, RIN limited	Modelled assuming MZM insertion loss reduced to only propagation loss of SiN _x (standard detection).
109.2	41.9, Shot noise limited	Modelled using differential detection with measured experimental conditions.
120.6	25.3, Shot noise limited	Modelled using differential detection and 1 W of optical power from laser source using all other measured link parameters.

4. Discussion and future work

The results reported in this manuscript demonstrate the capabilities of Sandia's heterogeneously integrated SiP/TFLN platform and provide framework for more complex integrated systems and higher performing travelling wave modulators. To substantially improve the electro-optic bandwidth of these devices, it will be necessary to simultaneously tune the phase velocity of the RF modes such that it is matched to the optical group velocity, utilize a CPW design such that attenuation is minimized while maintaining a characteristic impedance of 50 Ω. Designing a modulator which satisfies these three requirements is challenging, and the most straightforward way to realize these conditions are to change the design space which the MZM will ultimately occupy. For example, by utilizing a commercially available TFLN wafer with a silicon carrier wafer (as opposed to LiNbO₃), the RF phase velocity can be increased such that velocity matching is easier to implement, or, in other words, there can exist a wider design space that satisfies the requirement of velocity matching. By increasing the design space, there will exist more CPW designs that support the aforementioned conditions for bandwidth performance. To take full advantage of the design methodology presented here, CMOS compatible metal must be used, so the tuning of CPW propagation loss can only be realized through CPW design. In addition to using silicon as the carrier material for the TFLN wafer, one may also choose to remove the silicon material [8] to tune the phase velocity of the RF mode. Equivalently, the silicon handle wafer which serves as the substrate for the SiPIC may be removed through deep reactive ion etching (DRIE) [26]. While it adds an additional process step outside of normal CMOS-fabrication, it is not too cumbersome, and fabrication is still performed mostly using CMOS processes. CPWs are simulated using the methods discussed in this manuscript for a TFLN wafer utilizing a silicon

handle wafer and handle wafer removal, and it is found that the MZM reported in this work can exceed bandwidths of 40 GHz through removal of the TFLN carrier wafer (superstrate) using Eq. (3) and the simulated optical group index of 2.051. These results, along with improved frequency response using silicon as the superstrate material, are summarized in Fig. 10. Silicon is chosen as an alternative superstrate material due to its commercial availability. This illustrates how performance can be improved through implementation of different materials used in the heterogeneously integrated modulator, and it is worth mentioning that the results summarized in Fig. 10 correspond to the CPW geometries of the device reported in this manuscript. CPW with different geometries can be designed to optimize performance further for the case of TFLN carrier removal.

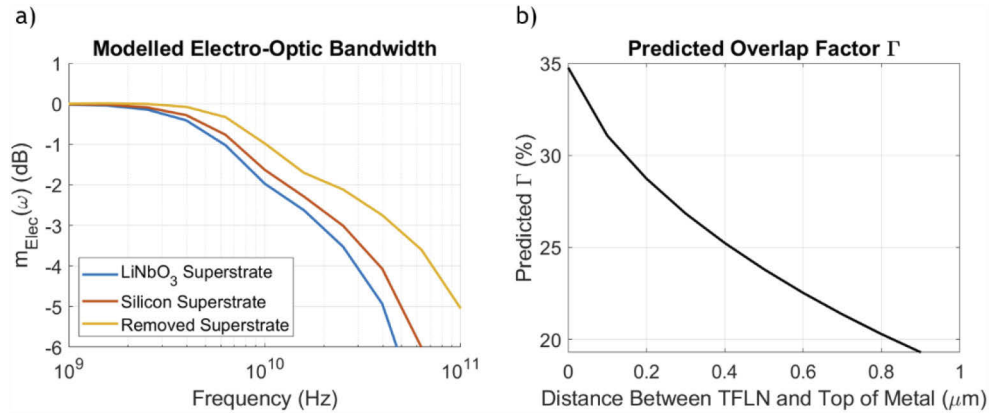


Fig. 10. (a) Predicted improvement in electro-optic frequency response of the reported modulator with TFLN on a silicon carrier wafer and removal of carrier wafer through reduction of velocity mismatch, and (b) the predicted improvement in overlap factor Γ as a function of distance between TFLN and the top of the CMOS compatible metal which is made possible through the use of damascene metallization.

The half-wave voltage may be improved through decreasing the distance between the TFLN and aluminum in an effort to increase overlap between RF and optical modes (Γ). To increase the value of Γ , a damascene metallization process could be utilized in which trenches would be etched into the oxide, filled with a CMOS compatible metal, and planarized such that the CPW electrodes would be formed. This approach is attractive because it is a CMOS compatible metallization technique [27] and removes the constraint of how far the RF waveguide can be placed from the modulation region. The predicted improvement in overlap factor is seen in Fig. 10 for a CPW that has a 4 μm gap between electrodes.

In summary, we have designed, fabricated, and characterized an electro-optic MZM which can entirely be manufactured in a CMOS foundry with the exception of a final bond of a TFLN sample. A path forward has been established towards ultimate device performance and maintain the attractive features of this platform such as removing the requirement to etch the LiNbO₃ and ability for integration with PICs. This is something that is not currently possible in pure TFLN platforms, in which modulators would need to be packaged and connectorized with fibers to be used.

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Disclosures

Sandia National Laboratories (P).

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